

Mapa de Modelos del Sistema

1. Modelos por Categoría

1.1 Modelos de Usuario

Modelo	Tabla	Relaciones
User	users	1:N → UserProgress, Subscription, Certificate
ChildProfile	child_profiles	N:1 → User
Subscription	subscriptions	N:1 → User

1.2 Modelos de Contenido

Modelo	Tabla	Relaciones
Unit	units	1:N → Topic, GrammarRule, VocabularyCategory, Flashcard
Topic	topics	N:1 → Unit
GrammarRule	grammar_rules	N:1 → Unit
VocabularyCategory	vocabulary_categories	N:1 → Unit, 1:N → VocabularyItem
VocabularyItem	vocabulary_items	N:1 → VocabularyCategory
Verb	verbs	1:N → VerbTense
VerbTense	verb_tenses	N:1 → Verb
Idiom	idioms	1:N → UserIdiomProgress
PhrasalVerb	phrasal_verbs	1:N → UserPhrasalVerbProgress
Reading	readings	1:N → ReadingQuestion
ReadingQuestion	reading_questions	N:1 → Reading
UnitExplanation	unit_explanations	N:1 → Unit
TopicExplanation	topic_explanations	N:1 → Topic

1.3 Modelos de Ejercicios

Modelo	Tabla	Relaciones
Quiz	quizzes	1:N → QuizQuestion
QuizQuestion	quiz_questions	N:1 → Quiz, 1:N → QuizOption

Modelo	Tabla	Relaciones
QuizOption	quiz_options	N:1 → QuizQuestion
SentenceExercise	sentence_exercises	1:N → UserSentenceExercise
UserSentenceExercise	user_sentence_exercises	N:1 → User, SentenceExercise
GrammarDrill	grammar_drills	1:N → UserDrillResult
UnitChallenge	unit_challenges	1:N → ChallengeQuestion
ChallengeQuestion	challenge_questions	N:1 → UnitChallenge
DailyChallenge	daily_challenges	1:N → UserDailyChallenge
QuickQuiz	quick_quizzes	1:N → UserQuizScore

1.4 Modelos de Progreso

Modelo	Tabla	Relaciones
UserProgress	user_progress	N:1 → User, Unit
UserGrammarProgress	user_grammar_progress	N:1 → User, GrammarRule
UserVocabularyProgress	user_vocabulary_progress	N:1 → User, VocabularyItem
StudyProgress	study_progress	N:1 → User
UserReadingScore	user_reading_scores	N:1 → User
UserQuizScore	user_quiz_scores	N:1 → User
ChildProgress	child_progress	N:1 → ChildProfile

1.5 Modelos de SRS

Modelo	Tabla	Relaciones
Flashcard	flashcards	1:N → UserFlashcardSRS, UserFlashcardReview
UserFlashcardSRS	user_flashcard_srs	N:1 → User, Flashcard
UserFlashcardReview	user_flashcard_reviews	N:1 → User, Flashcard

1.6 Modelos de Gamificación

Modelo	Tabla	Relaciones
Badge	badges	N:N → User (via user_badges)
UserStreak	user_streaks	N:1 → User

Modelo	Tabla	Relaciones
UserPoints	user_points	N:1 → User
PointsTransaction	points_transactions	N:1 → User
MiniGame	mini_games	1:N → MiniGameContent, UserGameScore
MiniGameContent	mini_game_content	N:1 → MiniGame
UserGameScore	user_game_scores	N:1 → User, MiniGame

1.7 Modelos de Exámenes

Modelo	Tabla	Relaciones
ExamSimulator	exam_simulators	1:N → ExamSection
ExamSection	exam_sections	N:1 → ExamSimulator
UserExamAttempt	user_exam_attempts	N:1 → User, ExamSimulator

1.8 Modelos de Conversación

Modelo	Tabla	Relaciones
Conversation	conversations	1:N → ConversationLine
ConversationLine	conversation_lines	N:1 → Conversation
ConversationPractice	conversation_practice	N:1 → User, Conversation
AlternativeResponse	alternative_responses	N:1 → ConversationLine
ResponsePattern	response_patterns	N:1 → ConversationLine
UserSentence	user_sentences	N:1 → User
SentenceLike	sentence_likes	N:1 → UserSentence

1.9 Modelos de Escenarios

Modelo	Tabla	Relaciones
ThematicScenario	thematic_scenarios	N:N → User (via user_unlocked_scenarios)
ScenarioVocabulary	scenario_vocabulary	N:1 → ThematicScenario
ScenarioPhrase	scenario_phrases	N:1 → ThematicScenario
ScenarioSimulation	scenario_simulations	N:1 → ThematicScenario
UserScenarioProgress	user_scenario_progress	N:1 → User, ThematicScenario

1.10 Modelos de Certificados

Modelo	Tabla	Relaciones
Certificate	certificates	N:1 → User

1.11 Modelos de Feedback

Modelo	Tabla	Relaciones
UserFeedback	user_feedback	N:1 → User
UserActivity	user_activities	N:1 → User
Bookmark	bookmarks	N:1 → User

1.12 Modelos de Análisis

Modelo	Tabla	Relaciones
WritingPractice	writing_practice	N:1 → User
UserWritingSubmission	user_writing_submissions	N:1 → User
UserReadingSubmission	user_reading_submissions	N:1 → User, Reading
UserSentencePractice	user_sentence_practice	N:1 → User
QuizSubmission	quiz_submissions	N:1 → User, Quiz
UserChallengeAttempt	user_challenge_attempts	N:1 → User, UnitChallenge
UserDailyChallenge	user_daily_challenges	N:1 → User, DailyChallenge
UserTypingScore	user_typing_scores	N:1 → User
WritingAnalysisLog	writing_analysis_logs	N:1 → User
ReviewSessionLog	review_session_logs	N:1 → User
GrammarExerciseResult	grammar_exercise_results	N:1 → User
ErrorLog	error_logs	N:1 → User
UserErrorPattern	user_error_patterns	N:1 → User

1.13 Modelos de Sistema

Modelo	Tabla	Relaciones
MotivationalMessage	motivational_messages	-
AudioFile	audio_files	-

1.14 Modelos de Licencias (Business)

Modelo	Tabla	Relaciones
RestaurantLicense	restaurant_licenses	N:1 → User
RestaurantEmployee	restaurant_employees	N:1 → RestaurantLicense

2. Tablas de Asociación

Tabla	Relaciones Many-to-Many
user_badges	User ↔ Badge
user_unlocked_scenarios	User ↔ ThematicScenario

3. Resumen de Relaciones

One-to-Many (1:N)

- User → UserProgress
- User → Subscription
- User → Certificate
- User → ChildProfile
- User → UserStreak
- User → Badge
- Unit → Topic
- Unit → GrammarRule
- Unit → VocabularyCategory
- Unit → Flashcard
- VocabularyCategory → VocabularyItem
- Verb → VerbTense
- Quiz → QuizQuestion
- QuizQuestion → QuizOption
- Conversation → ConversationLine
- ExamSimulator → ExamSection

Many-to-Many (N:N)

- User ↔ Badge (via user_badges)
- User ↔ ThematicScenario (via user_unlocked_scenarios)

4. Modelos con Campos JSON

Modelo	Campo JSON
Unit	learning_objectives

Modelo	Campo JSON
UserProgress	completed_topics
Badge	criteria
MiniGameContent	content
UserActivity	metadata
StudyProgress	completed_items

Mapa de modelos - English Learning Platform